

# Relations amongst security notions

Sebastian David Albu, Modern Public Key Cryptography 2024

May 15, 2024

## Related Work

This presentation is entirely based on the publication by “Relations amongst security notions”, from 2001

<https://cseweb.ucsd.edu/~mihir/papers/relations.pdf>

# Security Notions of Encryption Scheme Security

We consider three different attacks. In order of increasing strength:

- Chosen Plaintext Attack (CPA)
- Non-adaptive Chosen Ciphertext Attack (CCA-1)
- Adaptive Chosen Ciphertext Attack (CCA-2)

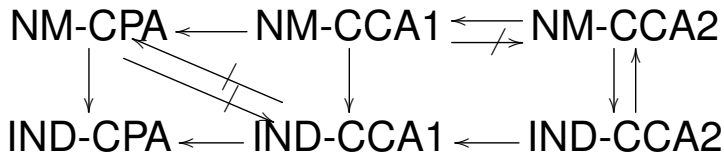
# Security Notions of Encryption Scheme Security

These can however also be mixed and matched with *Indistinguishability* (*IND*) and *Non-Malleability* (*NM*), resulting in 6 total security notions:

- IND-CPA and NM-CPA
- IND-CCA1 and NM-CCA1
- IND-CCA2 and NM-CCA2

The aim of this presentation is to relate these notions now.

# Implications and Separations



- $\mathbf{A} \Rightarrow \mathbf{B}$ : A proof that if  $\mathcal{PE}$  is any encryption scheme meeting notion of security  $\mathbf{A}$  then  $\mathcal{PE}$  also meets notion of security  $\mathbf{B}$ .
- $\mathbf{A} \not\Rightarrow \mathbf{B}$ : A construction of an encryption scheme  $\mathcal{PE}$  that provably meets notion of security  $\mathbf{A}$  but provably does *not* meet notion of security  $\mathbf{B}$ .<sup>2</sup>

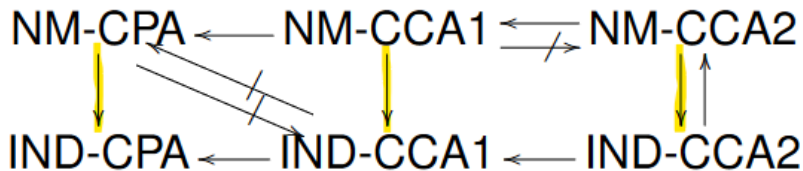
## Framework

Let  $A = (A_1, A_2)$  be any adversary attacking encryption scheme  $\mathcal{PE}$  in the sense of IND-ATK/NM-ATK. Then there exists another adversary  $B = (B_1, B_2)$  attacking  $\mathcal{PE}$  in the sense of IND-ATK/NM-ATK such that the two (equal length) messages that  $B_1$  outputs are always distinct,

$$\text{Adv}_{\mathcal{PE}, B}^{\text{IND-ATK/NM-ATK}}(k) = \text{Adv}_{\mathcal{PE}, A}^{\text{IND-ATK/NM-ATK}}(k),$$

and the running time of  $B$  is within a constant factor of that of  $A$ .

# Non-Malleability (atk) implies Indistinguishability (atk)



$$\text{NM-atk} \rightarrow \text{IND-atk}$$

## Proof $\text{NM} \rightarrow \text{IND}$

Adversaries  $A$  and  $B$  have access to an oracle  $\mathcal{O}^1$  in the first stage and to another oracle  $\mathcal{O}^2$  in the second stage.

Algorithm  $A_1^{\mathcal{O}^1}(pk)$   
 $(x_0, x_1, s) \leftarrow B_1^{\mathcal{O}^1}(pk)$   
 $M := \{x_0, x_1\}$   
 $s' \leftarrow (x_0, x_1, pk, s)$   
**return**  $(M, s')$

Algorithm  $A_2^{\mathcal{O}^2}(M, s', y)$  where  $s' = (x_0, x_1, pk, s)$   
 $c \leftarrow B_2^{\mathcal{O}^2}(x_0, x_1, s, y)$   
 $y' \leftarrow \mathcal{E}_{pk}(\overline{x_c})$   
**return**  $(R, y')$  where  $R(a, b) = 1$  iff  $a = \overline{b}$

# Proof $\text{NM} \rightarrow \text{IND}$ - The advantages of $A$

$$\text{Adv}_{\mathcal{PE}, A}^{\text{nm-atk}}(k) = \Pr[\text{Exp}_{\mathcal{PE}, A}^{\text{nm-atk-1}}(k) = 1] - \Pr[\text{Exp}_{\mathcal{PE}, A}^{\text{nm-atk-0}}(k) = 1]$$

with

$$\begin{aligned} \text{Exp}_{\mathcal{PE}, A}^{\text{nm-atk-1}}(k) &\stackrel{\text{def}}{=} \left[ (pk, sk) \leftarrow \mathcal{K}(k) ; (M, s') \leftarrow A_1^{\mathcal{O}_1}(pk) ; x \leftarrow M ; y \leftarrow \mathcal{E}_{pk}(x) ; \right. \\ &\quad \left. (R, y') \leftarrow A_2^{\mathcal{O}_2}(M, s', y) ; x' \leftarrow \mathcal{D}_{sk}(y') : y \neq y' \wedge \perp \neq x' \wedge R(x, x') \right] \\ \text{Exp}_{\mathcal{PE}, A}^{\text{nm-atk-0}}(k) &\stackrel{\text{def}}{=} \left[ (pk, sk) \leftarrow \mathcal{K}(k) ; (M, s') \leftarrow A_1^{\mathcal{O}_1}(pk) ; x, \tilde{x} \leftarrow M ; y \leftarrow \mathcal{E}_{pk}(x) ; \right. \\ &\quad \left. (R, y') \leftarrow A_2^{\mathcal{O}_2}(M, s', y) ; x' \leftarrow \mathcal{D}_{sk}(y') : y \neq y' \wedge \perp \neq x' \wedge R(\tilde{x}, x') \right] . \end{aligned}$$

## Proof NM $\rightarrow$ IND - The advantages of $A$

$$\mathit{Adv}_{\mathcal{P}\mathcal{E},A}^{\text{nm-atk}}(k) = \Pr[\text{Exp}_{\mathcal{P}\mathcal{E},A}^{\text{nm-atk-1}}(k) = 1] - \Pr[\text{Exp}_{\mathcal{P}\mathcal{E},A}^{\text{nm-atk-0}}(k) = 1]$$

$A$  obtains information from  $\mathcal{O}^1$  to get ciphertexts.

$A$  interacts with  $\mathcal{O}^2$  to obtain modified decryptions of the ciphertexts.

$\rightarrow A$  tries to attack malleability

Proof NM  $\rightarrow$  IND - The advantage of  $B$ 

$$Adv_{\mathcal{PE}, B}^{\text{ind-atk}}(k) = 2 \cdot \Pr[\text{Exp}_{\mathcal{PE}, B}^{\text{ind-atk-b}}(k) = b] - 1$$

with

$$\text{Exp}_{\mathcal{PE}, B}^{\text{ind-atk-b}}(k) \stackrel{\text{def}}{=} \Pr \left[ (pk, sk) \leftarrow \mathcal{K}(k) ; (x_0, x_1, s) \leftarrow B_1^{\mathcal{O}_1}(pk) ; b \leftarrow \{0, 1\} ; \right. \\ \left. y \leftarrow \mathcal{E}_{pk}(x_b) ; c \leftarrow B_2^{\mathcal{O}_2}(x_0, x_1, s, y) : c = b \right].$$

## Proof NM $\rightarrow$ IND - Summary Advantages

- Adversary  $A$  targets non-malleability (NM-ATK) and tries to exploit any potential malleability in the encryption scheme  $\mathcal{PE}$ .
- Adversary  $B$  targets indistinguishability (IND-ATK) and tries to exploit any patterns or weaknesses that might allow it to distinguish between encryptions of different messages.

## Proof NM $\rightarrow$ IND - First Claim and Proof

Claim:

$$\Pr[\text{Exp}_{\mathcal{P}\mathcal{E},A}^{\text{nm-atk-1}}(k) = 1] = \Pr[\text{Exp}_{\mathcal{P}\mathcal{E},B}^{\text{ind-atk-b}}(k) = b]$$

Proof:

The proof relies on the observation that the condition  $R(x, x_0)$  in  $\text{Exp}_{\mathcal{P}\mathcal{E},A}^{\text{nm-atk-1}}$  is equivalent to  $b = c$  in  $\text{Exp}_{\mathcal{P}\mathcal{E},B}^{\text{ind-atk-b}}$ , where  $b$  and  $c$  represent the bits used to select the plaintext in the encryption process.  $b = c$  in the experiment underlying  $pk$  exactly when  $y \neq y_0 \wedge \perp \neq x_0 \wedge R(x, x_0)$  in  $\text{Exp}_{\mathcal{P}\mathcal{E},A}^{\text{nm-atk-1}}(k)$ .

## Proof NM $\rightarrow$ IND - Second Claim and Proof

Claim:

$$\Pr[\text{Exp}_{\mathcal{PE}, A}^{\text{nm-atk-0}}(k) = 1] = \frac{1}{2}$$

Proof:

Relies on information theoretical fact:

$A$  has no information about the message  $\tilde{x}$  with respect to which its success is measured.

## Proof $\text{NM} \rightarrow \text{IND}$ - Conclusion

Apply claims to get  $\text{Adv}_{\mathcal{PE},B}^{\text{ind-atk}}(k) = 2 \cdot \text{Adv}_{\mathcal{PE},A}^{\text{nm-atk}}(k)$ .

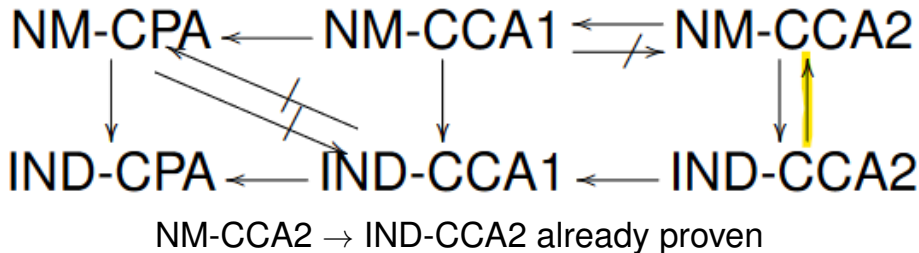
—

Since  $\mathcal{PE}$  is secure in the NM-ATK sense,  $\text{Adv}_{\mathcal{PE},A}^{\text{nm-atk}}(\cdot)$  is negligible.

—

This implies  $\text{Adv}_{\mathcal{PE},B}^{\text{ind-atk}}(\cdot)$  is negligible too. □

## IND-CCA2 and NM-CCA2



# IND-CCA2 $\rightarrow$ NM-CCA2

Intuition:

Since the adversary has access to the decryption oracle, they can decrypt the ciphertexts they would output, and so the ability to output ciphertexts is not likely to add power

## Proof $\text{IND-CCA2} \rightarrow \text{NM-CCA2}$

$B = (B_1, B_2)$  is NM-CCA2 adversary attacking  $\mathcal{PE}$

$A = (A_1, A_2)$  is an IND-CCA2 adversary

—

To be proven:  $\text{Adv}_{\mathcal{PE}, B}^{\text{nm-cca2}}(\cdot)$  is negligible

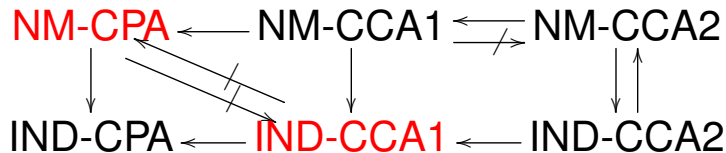
## Proof IND-CCA2 $\rightarrow$ NM-CCA2

The intuition is already quite obvious

Crafting and relating  $Adv_{\mathcal{P}\mathcal{E},B}^{nm-cca2}(\cdot)$  and  $Adv_{\mathcal{P}\mathcal{E},A}^{ind-cca2}(\cdot)$  reveals:

$$Adv_{\mathcal{P}\mathcal{E},B}^{nm-cca2}(\cdot) = 2 \cdot Adv_{\mathcal{P}\mathcal{E},A}^{ind-cca2}(\cdot) \quad \square$$

## NM-CPA and IND-CCA1



# Proof IND-CCA1 $\not\rightarrow$ NM-CPA

$\mathcal{PE}'$  which is IND-CCA1, but not NM-CPA

—

|                                                                                          |                                                                                                                                              |                                                                             |
|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Algorithm $\mathcal{K}'(k)$<br>$(pk, sk) \leftarrow \mathcal{K}(k)$<br>return $(pk, sk)$ | Algorithm $\mathcal{E}'_{pk}(x)$<br>$y_1 \leftarrow \mathcal{E}_{pk}(x)$ ; $y_2 \leftarrow \mathcal{E}_{pk}(\bar{x})$<br>return $y_1    y_2$ | Algorithm $\mathcal{D}'_{sk}(y_1    y_2)$<br>return $\mathcal{D}_{sk}(y_1)$ |
|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|

# Proof IND-CCA1 $\not\rightarrow$ NM-CPA

Claim:

$\mathcal{PE}'$  is not NM-CPA

—

Proof:

Given ciphertext  $y_1 || y_2$  of message  $x$ , it is easy to craft a ciphertext of  $\bar{x}$ :  $y_2 || y_1$ , since  $\mathcal{D}'_{sk}(y_2 || y_1) = \mathcal{D}_{sk}(y_2)$ .

# Proof IND-CCA1 $\not\rightarrow$ NM-CPA

Claim:

$\mathcal{PE}'$  is IND-CCA1

Setup:

$B = (B_1, B_2)$  attacker in IND-CCA1 sense.

Proof: Show  $Adv_{\mathcal{PE}', B}^{ind-cca1}(\cdot)$  negligible

# Proof IND-CCA1 $\not\rightarrow$ NM-CPA

For  $i, j \in \{0, 1\}$

$$p_k(i, j) = \Pr \left[ (pk, sk) \leftarrow \mathcal{K}(k) ; (x_0, x_1, s) \leftarrow B_1^{\mathcal{D}_{sk}}(pk) ; y_1 \leftarrow \mathcal{E}_{pk}(x_i) ; y_2 \leftarrow \mathcal{E}_{pk}(\overline{x_j}) : B_2(x_0, x_1, s, y_1 \| y_2) = 1 \right] .$$

$$Adv_{\mathcal{P}_{\mathcal{E}', B}}^{ind-cca1}(k) = p_k(1, 1) - p_k(0, 0)$$

## Proof IND-CCA1 $\not\rightarrow$ NM-CPA

To complete the proof:  $p_k(1, 1) - p_k(1, 0)$  and  $p_k(1, 0) - p_k(0, 0)$  both negligible.

This works because

$$Adv_{\mathcal{PE}', B}^{ind-cca1}(k) = p_k(1, 1) - p_k(0, 0) =$$

$$[p_k(1, 1) - p_k(1, 0)] + [p_k(1, 0) - p_k(0, 0)] = p_k(1, 1) - p_k(0, 0)$$

# Proof $\text{IND-CCA1} \not\rightarrow \text{NM-CPA}$

Proof  $p_k(1, 1) - p_k(1, 0)$  is negligible

—

Setup:

Algorithm  $A_1^{\mathcal{D}_{sk}}(pk)$   
 $(x_0, x_1, s) \leftarrow B_1^{\mathcal{D}'_{sk}}(pk)$   
 $m_0 \leftarrow \overline{x_0}; m_1 \leftarrow \overline{x_1}$   
**return**  $(m_0, m_1, s)$

Algorithm  $A_2(m_0, m_1, s, y)$   
 $y_1 \leftarrow \mathcal{E}_{pk}(\overline{m_1}); y_2 \leftarrow y$   
 $d \leftarrow B_2(\overline{m_0}, \overline{m_1}, s, y_1 \parallel y_2)$   
**return**  $d$

## Proof IND-CCA1 $\nrightarrow$ NM-CPA

The computation  $B_1^{\mathcal{D}'_{sk}}(pk)$  is done by  $A_1$  simulating the  $\mathcal{D}'_{sk}$  oracle. It can do this by replying to query  $y_1 || y_2$  via  $\mathcal{D}_{sk}(y_1)$ , using its own  $\mathcal{D}_{sk}$  oracle and the definition of  $\mathcal{D}'_{sk}$ .

$$Adv_{\mathcal{PE}', B}^{ind-cca1}(k) = p_k(1, 1) - p_k(1, 0) \quad \square$$

—

For  $Adv_{\mathcal{PE}', B}^{ind-cca1}(k) = p_k(1, 1) - p_k(1, 0)$  same proof and similar setup.

# Proof IND-CCA1 $\not\rightarrow$ NM-CPA

*Proof:* We can construct an adversary  $A = (A_1, A_2)$  that attacks the scheme  $\mathcal{PE}$  in the IND-CCA1 sense, as follows:

|                                                                                                                                                                                      |  |                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Algorithm</b> <math>A_1^{\mathcal{D}_{sk}}(pk)</math><br/> <math>(x_0, x_1, s) \leftarrow B_1^{\mathcal{D}'_{sk}}(pk)</math><br/> <b>return</b> <math>(x_0, x_1, s)</math></p> |  | <p><b>Algorithm</b> <math>A_2(x_0, x_1, s, y)</math><br/> <math>y_1 \leftarrow y</math> and <math>y_2 \leftarrow \mathcal{E}_{pk}(\overline{x_0})</math><br/> <math>d \leftarrow B_2(x_0, x_1, s, y_1 \  y_2)</math><br/> <b>return</b> <math>d</math></p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Again  $A$  is polynomial time and can simulate  $\mathcal{D}'_{sk}$  given  $\mathcal{D}_{sk}$ . We observe that

$$\Pr \left[ (pk, sk) \leftarrow \mathcal{K}(k) ; (x_0, x_1, s) \leftarrow A_1^{\mathcal{D}_{sk}}(pk) ; y \leftarrow \mathcal{E}_{pk}(x_1) : A_2(x_0, x_1, s, y) = 1 \right] = p_k(1, 0)$$

$$\Pr \left[ (pk, sk) \leftarrow \mathcal{K}(k) ; (x_0, x_1, s) \leftarrow A_1^{\mathcal{D}_{sk}}(pk) ; y \leftarrow \mathcal{E}_{pk}(x_0) : A_2(x_0, x_1, s, y) = 1 \right] = p_k(0, 0)$$

Thus  $\text{Adv}_{\mathcal{PE}, A}^{\text{ind-cca1}}(k) = p_k(1, 0) - p_k(0, 0)$ . The assumed security of  $\mathcal{PE}$  in the IND-CCA1 sense now implies the latter difference is negligible.  $\square$

# IND-CCA1 $\rightarrow$ NM-CCA1 (?)

As a fun side note:

*“It is interesting from a theoretical standpoint that the equivalence holds for this relaxed version of IND-CCA2 and NM-CCA2, whereas it is known that IND-CCA1 does not imply NM-CCA1. [7]”*

# IND-CCA1 $\rightarrow$ NM-CCA1 (?)

7. Bellare, M., Desai, A., Pointcheval, D., Rogaway, P.: Relations among notions of security for public-key encryption schemes. In: CRYPTO'98. pp. 26–45. Springer (1998)

They link the original paper which we currently summarize!

## Further Reading

For the other proofs from the shown picture, refer to the original paper:

<https://cseweb.ucsd.edu/~mihir/papers/relations.pdf>

—

Google/First result when searching IND-CCA1 ➔ NM-CCA1:

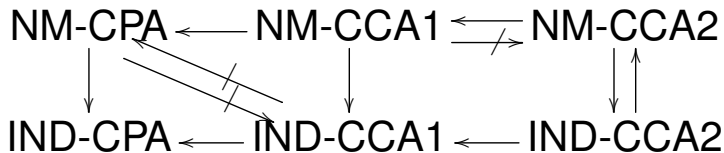
<https://eprint.iacr.org/2024/202.pdf>

(They use the original paper discussed as a reference)

Circular dependency :)

## Conclusion

# Conclusion



To be continued (?) :)